

Who Gets to See the Map?

AI, shared situational awareness, and why information in organizations flows up but not down

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AI could give every employee the same real-time picture the executive team sees. In most companies, that picture still ends up in a monthly deck. The bottleneck isn't technology. It's leadership.

On January 4, 1940, in a frozen forest in eastern Finland, a Soviet division commander named Aleksei Vinogradov radioed his army headquarters with a simple request. His 44th Division, a motorized unit of some 17,000 men, had been chopped into isolated pockets by Finnish ski troops. Vinogradov could see what was happening. He asked for permission to act on his own judgment. Headquarters said no. By the time permission to withdraw came two days later, there was very little division left to withdraw. A week after that, Vinogradov was executed by firing squad in front of the survivors. Few executives today face that kind of consequence for a bad quarter. But almost every leader knows the underlying situation: waiting for approval from people who are further from the facts than you are.

Some context for readers who don't know the story. In the winter of 1939–40, the Soviet Union invaded Finland, a country of under four million people, expecting a war of a few weeks. The Finns held out for more than three months. One reason was a tactic that nobody had written into the field manuals before the war. Small Finnish units learned on the spot to cut road-bound Soviet columns into isolated pockets, which they called *motti*, the Finnish word for a measured stack of firewood, and then destroy them one by one. The tactic spread sideways, from unit to unit, before headquarters made it official. On one side, an army where the people closest to the problem could act. On the other, an army where initiative could get you shot.

The Monday meeting where everyone already knew

Fast forward to a modern plant. The production line carries hundreds of sensors, the ERP system logs every transaction, and maintenance can see a bearing temperature to the second. On the first Monday of the month, the leadership team learns from a slide that line efficiency dipped in the second week of the previous month. The operators knew that at the time. But the information traveled first into a system, then to a controller, then into a deck, and finally

into a conference room. If it ever makes it back to the line, it arrives as an improvement project. None of this is scandalous. It is simply how most companies still run. We collect more data than ever, and the technology to show it to everyone in near real time is cheap and mature. Yet data still flows up and decisions still flow down, and both directions are slow. The PowerPoint deck remains one of the slowest communication technologies ever devised, narrowly ahead of the carrier pigeon.

Ukraine's shared map

Ukraine's military has had to operate differently because it cannot afford the delay. Its Delta system fuses feeds from drones, satellites, sensors and reconnaissance units onto a single map that is visible both at headquarters and on a soldier's phone. Access is tiered, but the map is the same. An artillery app assigns fire missions to whichever gun is available, much the way a ride-hailing app assigns trips. Drone units earn points for video-verified strikes and spend them in an online marketplace for new equipment instead of waiting for an allocation from above.

It isn't a clean success story. Ukrainian commanders have been criticized for micromanagement, and the points system tempts units to chase high-scoring targets rather than the ones that matter most. A researcher at West Point's Modern War Institute put the problem well this summer: the battlefield can become more transparent without becoming more understandable. A brigade commander watching live drone footage now sees what only a platoon leader used to see, and the temptation to take over the platoon leader's job grows accordingly. The same technology that pushes information to the edge also pulls the edge onto the general's screen.

The org chart was born from a data flood

The organizational chart was invented in the 1850s by Daniel McCallum, a superintendent of the New York and Erie Railroad. The telegraph was delivering more operational data than anyone could process, and his chart was a way to route it: who reports to whom, and who decides what. Middle management was created to move information, not primarily to command.

That history has suddenly become relevant again. In March, Block CEO Jack Dorsey and Sequoia's Roelof Botha published an essay arguing that corporate hierarchy is an information-routing protocol dating back to the Roman legions, and that AI can now replace it. In their model, AI maintains a continuously updated "world model" of the company: what is being built, what is blocked, where resources are going. A permanent middle-management layer becomes unnecessary, and people work at the edge where the system meets reality. Block had just cut roughly 40 percent of its workforce, and analysts were quick to point out that much of that reflected pandemic-era overhiring rather than AI. The essay is still worth reading. The diagnosis is sharp, even if the prescription is debatable.

A Finnish answer: hierarchy doesn't disappear, it changes its job

Finland offers a useful counterpoint. In the spring of 1943, during Finland's second war with the Soviet Union, an instructor at the Finnish army's artillery school, Major Unto Petäjä, developed a simple plotting device known as the correction converter. It meant a forward observer no longer needed to know where the guns were. The math moved to the firing positions, and the observer could call corrections as if every gun were standing right next to him. In the summer of 1944, at Tali-Ihantala, one of the largest battles ever fought in the Nordic countries, Finnish forward observers were able to bring down the fire of 254 guns on a single target. A piece of plywood did 80 years ago what we now expect from AI: it removed the calculating middle layer and gave the person with eyes on the problem access to the whole system.

But the hierarchy didn't vanish. The largest concentrations still required approval from above. The edge got the means; the center kept the authorization. The same pattern shows up in some of Finland's best-known modern organizations. At Supercell, the Helsinki game studio behind Clash of Clans, small game teams decide for themselves whether to continue or kill a game, but success metrics are agreed in advance and shared with the whole company. Linux, created by the Finnish engineer Linus Torvalds, has exactly one person who can merge code into the main kernel, yet he personally picks only a tiny fraction of the changes and relies on a chain of trusted subsystem maintainers. The project's own documentation compares being a maintainer to an on-call rotation rather than a position of power.

So in the age of AI, hierarchy doesn't disappear. It changes its job. The command hierarchy, which pulls information up and pushes decisions down, becomes obsolete. What replaces it is a hierarchy of permission, which gives the front line the means and keeps a few big calls at the top, and a hierarchy of trust, which guarantees quality and owns its territory without issuing orders.

Fear travels faster than fiber

If technology were enough, information would already be flowing. Research on Nokia, the Finnish company that dominated the global mobile phone market before the iPhone, explains why it isn't. Timo Vuori of Aalto University and Quy Huy of INSEAD found that Nokia's top managers feared competitors and shareholders, while middle managers feared top management. Middle managers stopped delivering bad news, and top management ended up with an overly rosy picture of the company's technical capabilities. The information existed inside the company. It just didn't move.

AI doesn't remove that mechanism. It gives it a new form. Wharton professor Ethan Mollick has described "secret cyborgs": the most skilled AI users in many organizations hide their use, partly because unclear policies make them fear punishment and partly because revealing a big productivity jump could put their own or a colleague's job at risk. At Nokia, people hoarded bad news. In the AI era, they hoard good news.

The camera is pointing the wrong way

When data is shared, direction matters. Uber knows almost everything about its drivers, yet researchers found that drivers long had to accept ride requests without seeing the fare. The system knew; the worker didn't. Ukraine's Delta works the other way. The technology is similar. The choice is different.

Harvard's Ethan Bernstein studied a large mobile phone factory in China and found that constant observation reduced performance, because workers started hiding what they were doing. When some lines were screened off with curtains, their productivity improved and stayed higher. The idea for the curtains came from a worker, and output measurement continued the whole time. A field experiment at a U.S. trucking company found that posting named driver rankings improved performance at sites with an individualistic culture but hurt it at sites that had just begun building a more collaborative one. Management dropped the rankings as soon as it saw the results.

The lesson is simple. Data *about* workers, sent up to management, easily undermines performance. Data *for* workers, about their own work and in the right form, improves it. Toyota understood this in the 1950s: one of the principles of its production system is that neither goods nor information should be held up along the way, and problems become visible first where they occur.

When we turned the numbers over to the shop floor

I have seen this firsthand. In a plant I ran, we had spent years trying to reduce scrap costs. The approach was familiar: the quality organization analyzed the data, and management assigned projects to bring scrap down. People worked hard, but the results barely moved. We flipped the model. Real-time quality data went straight to the shop floor, and the analysis told the people doing the work how big each problem was and which way it was trending. Production formed its own teams, chose its own projects, and agreed among themselves who would do what and by when.

Management's role changed from giving orders to listening. We met the teams once a month, heard their plans, and asked what support they needed to carry them out. Within a few months the improvement was visible in the numbers, and after a year the result was dramatic. The data was the same as before, and so were the people. The only things that changed were who got to see the map and who got to choose the route.

On the factory floor, the world model is partly blind

Dorsey's model works in a remote-first company where all work leaves a digital trace. A factory is different. A shift supervisor's instinct, a maintenance technician's ear and the mood on the line don't flow into any system. In manufacturing, the world model is partly blind, which makes the person at the edge even more important than in Silicon Valley.

Swedish fintech Klarna learned a version of this in customer service. It announced that an AI assistant was doing the work of hundreds of agents, then had to bring humans back when service quality slipped; its CEO acknowledged the company had focused too much on cost.

Klarna tracked conversation volume and cost closely but did not track customer outcomes with the same rigor. When a machine optimizes a metric, the metric had better be the thing you actually want.

What's left for leaders

AI removes the two classic excuses for keeping data at the top. Interpretation no longer requires an analyst, because a model can summarize data for every role. Distribution no longer requires a report, because information can go straight to a phone, a chat or a machine display. What remains are power, fear and habit, which is to say, leadership choices.

That leaves leaders with three jobs machines won't do. The first is defining purpose, because AI agents execute well-defined goals brilliantly and vague ones poorly. The second is named accountability: intelligence may scale, but accountability doesn't, so every piece of work done by agents needs a human who answers for it by name. The third is psychological safety, because without it your best discoveries stay in someone's drawer.

Finland has a head start here that few countries share. Its culture has unusually low power distance, the Finnish Defence Forces build their doctrine on mission command, and because Finland still has broad conscription, more than 100,000 young Finns have gone through the army's trust-based leadership training as conscript leaders. Even so, Finnish business has more often borrowed the army's discipline than its mission command. Vinogradov's radio message was a request for permission to think. In the age of AI, every leader should ask how many people in their own organization are still waiting for that permission.

Questions for your organization

- How long does it take for a deviation in production, sales or service to reach the person who can fix it? How long to reach the executive team?
- How much of the data you collect comes back to the front line in usable form, and how much only feeds reports that go up?
- Does a frontline supervisor see the same picture as the CEO? If not, who decided that, and why?
- Where does the front line have to wait for approval even though it has the best information?
- Who in your organization uses AI most skillfully, and does leadership know? Would that person feel safe telling you?
- When did someone last deliver bad news to the top without it being softened on the way?
- Do your AI initiatives measure only cost and volume, or also outcomes for the customer?
- Whose name is accountable for what each AI agent does?

What to try

- **Make information latency a KPI.** Track how quickly data gets back to the people doing the work, with the same seriousness as cycle time.
- **Build one map for everyone.** Design the operational picture for the front line first and the executive report second. Tier access if you must, but don't split the map.
- **Send data to people, not just about them.** Give everyone the data on their own work. Think hard before publishing named rankings.
- **Set the kill criteria up front.** Agree on what success looks like before the team falls in love with the project, and share it with everyone.
- **Move from command to permission.** List the decisions that require approval from above and push the rest to where the information originates.
- **Reward sharing.** Make it safe and rewarding to share AI breakthroughs so hidden expertise comes to the surface.
- **Keep humans at the edge.** Your factory's world model can't see everything. An experienced operator is a sensor no system can replace.

Sources

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